AI Support Chatbot

Deployment & Strategy Overview

Avis · Budget · Self-hosted · Private · RAG-powered

Version	1.0 — Draft for internal review
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Architecture	Dual RAG + Local LLM
Privacy	100% on-premise
Model	LLaMA 3.3 70B via Ollama
Audience	Leadership and management

1. What Is This System?

This project is an internal AI assistant — a chatbot your call center agents can ask questions to, and it will answer based on your own documentation and past resolved cases. It works like a knowledgeable colleague who has read every page of your manual and remembers every case your team has ever solved.

> **No data leaves your organization.** This system runs entirely on your own infrastructure. Nothing is ever sent to Google, OpenAI, Microsoft, or any external company.

1.1 What it can do

— Answer questions about procedures, policies, and guidelines from your internal documentation
— Find and reference similar past cases to help agents resolve current situations faster
— Give consistent answers based on official documentation — not from memory or personal opinion
— Support multiple agents simultaneously without performance issues
— Be updated instantly when new documentation or resolved cases are added

1.2 What it cannot do

— Does not connect to the internet or any external service
— Only knows what you give it — if the answer is not in your documentation, it will say so
— Does not replace your agents' judgment — it helps them work faster and more consistently
— Not a general-purpose AI — focused exclusively on your knowledge base

2. Expected Impact — Key Performance Metrics

The following projections are based on industry benchmarks for call centers that have implemented AI-assisted knowledge retrieval tools. Conservative targets achievable within the first 6 months of full deployment.

Objective	Current Estimate	Target with Chatbot
Time to find information during a call	~2–3 minutes per call	**~15–30 seconds**
Average Handle Time (AHT)	~8–10 minutes	**~5–7 minutes**
First Call Resolution (FCR)	~70%	**~82–85%**
Escalations to supervisor	~25–30% of calls	**~12–15% of calls**

Objective	Current Estimate	Target with Chatbot
Onboarding time for new agents	4–6 weeks	**2–3 weeks**
Answer consistency across agents	Variable — depends on the agent	**High — all agents draw from the same source**
Agent experience on complex calls	Manual search under pressure	**Instant AI assistance available**

These figures are projections based on comparable call center deployments. A formal measurement baseline should be established before launch so results can be tracked accurately.

3. How Does It Work?

The system uses a technique called RAG — Retrieval-Augmented Generation. Before answering a question, the AI looks up the most relevant pages from your documentation and the most similar past cases, then uses those as context to write an accurate, grounded answer.

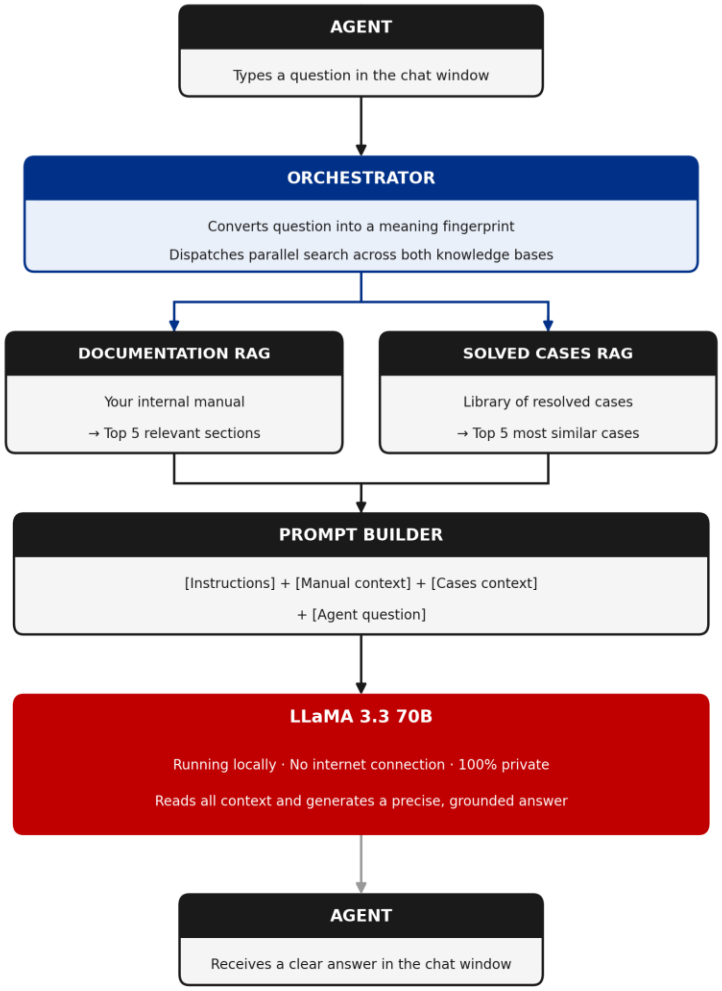
> **Why not just use ChatGPT?** Public AI tools do not know your procedures or cases and may give plausible but incorrect answers. This system answers exclusively from your verified knowledge base, grounded in your actual policies and precedents. Nothing your agents type is shared with any external company.

3.1 Step by step

Step	What happens
1	Agent asks a question — types their question into a chat window in their browser, like texting or using WhatsApp.
2	System searches the documentation — in milliseconds, finds the 5 most relevant sections of your documentation manual.
3	System searches the cases — simultaneously, finds the 5 most similar situations from your resolved case library.
4	AI reads the context and answers — reads all 10 pieces of context and writes a clear, accurate answer based exclusively on your knowledge base.
5	Answer appears in chat — the response appears in the chat window, usually within a few seconds.

3.2 Architecture diagram

Each block below is a component running on your server. Information flows from the agent's question, through two parallel searches, to the AI model, and back.



4. Deployment Options — Where Does the Chat Live?

The AI engine and knowledge base always run on your server — that part never changes. The question is how your agents access it. There are three options, and the right one depends on your timeline, budget, and whether you already have an internal application your agents use daily.

Option	Launch time	Cost	Best for	Requires
A — Ready-made chat page	1–2 days	Low	Fast rollout, separate link	No development
B — Inside your existing app	1–2 weeks	Medium	Seamless, context-aware	Frontend developer
C — Brand-new custom application	4–8 weeks	High	Full control, custom branding	Development team

Option A · Ready-made chat page

Agents open a link in their browser and use a chat interface that looks exactly like ChatGPT, with no development work required. They open a URL such as chat.yourcompany.com and are greeted with a clean, familiar chat interface. Nothing to install, no training required, works on any device with a browser.

— Ready to use in 1–2 days, with no development work needed

— Multiple user accounts and chat history built in, and works on any device with a browser
— Limitations: agents must switch to a separate window, the design cannot be customized, and the assistant does not know what the agent is currently working on

The chat interface software (Open WebUI) is free and open-source. The only cost is the server — see Section 5.

Option B · Embedded inside your existing application

Recommended for call center environments. The AI chat appears as a panel or button inside the application your agents already use when handling clients — no switching windows or tabs. Because the chat lives inside your existing application, it can automatically know which client, ticket, or case the agent is currently viewing, without the agent typing anything manually.

Approach	What it means	When to use it
Quick embed	The ready-made chat is embedded as a panel inside your existing app — like mounting a screen in the wall.	You want speed and your existing app can be modified slightly. Takes a few hours.
Native widget	A custom chat component built directly into your existing app, reading current client ID, ticket number, and live context automatically.	You want full context awareness and a seamless experience. Takes 1–2 weeks of developer work.

Option C · Brand-new custom application

Maximum control and flexibility: a fully custom-built web application designed from scratch with your branding, your features, and your workflow — including the AI chat plus client management, case tracking, dashboards, analytics, or any other tools your team needs. Highest cost, at 4–8 weeks of development, and requires a dedicated development team that owns full maintenance going forward.

5. Server & Hosting — Where Does the AI Run?

The AI model needs to run on a powerful server with specialized processors (GPUs) designed for AI workloads. Two main choices: a server in your own office, or a rented server in the cloud.

5.1 On-premise vs. cloud

	Your own server (on-premise)	Rented server (cloud)
Where the server lives	Inside your building	In a secure data center
Who manages hardware	Your IT team	The cloud provider
Upfront investment	High — purchase GPU hardware	None
Monthly cost	Electricity and maintenance only	Hourly rental fee while running
Does data leave the building?	Never	No — stays on your private server
Remote access	Office network only	Anywhere, with VPN
Time to launch	2–4 weeks (hardware delivery)	1–2 hours
Best for	Maximum privacy, high daily usage	Remote teams, fast start, flexibility

Is cloud safe for sensitive call center data? Yes. You rent a fully private machine — the cloud provider manages the physical hardware but has no visibility into your software, agent conversations, or client data. The server is configured so only your office network or VPN can connect.

5.2 Cloud provider options

Provider	Hardware	Cost / hour	Best for
AWS	A100 — 40 GB	~$3.20	Enterprise compliance certifications for regulated industries
Google Cloud	A100 — 40 GB	~$3.67	Teams already using Google Workspace
Azure	A100 — 80 GB	~$3.40	Microsoft 365 environments needing enterprise support
Lambda Labs	A100 — 40 GB	~$1.10	Budget-conscious teams; simple setup, fewer enterprise features
RunPod	A100 — 80 GB	~$1.64	Easy to get started; great for testing before committing

5.3 Estimated monthly cost

Option	Setup cost	Monthly cost	Notes
Own server (on-premise)	~$15,000–$30,000	~$100–$300	High upfront, very low ongoing
AWS — business hours only	None	~$480–$640	12 hrs/day, Mon–Fri. Storage ~$20/mo extra
AWS — 24 hours / 7 days	None	~$2,300	Recommended for overnight shifts
RunPod — business hours only	None	~$150–$250	Most affordable cloud option

> **Save 60–70% on cloud costs.** Schedule the server to turn on at 8am and off at 8pm every weekday automatically. An AWS server costing $2,300/month running 24/7 drops to $480–$640/month on a business-hours schedule.

5.4 Decision guide

Our situation is…	We should go with…
We want it running as fast as possible with minimal investment	Option A + RunPod or Lambda Labs cloud
We have an existing app our agents use daily and want the AI inside it	Option B + cloud or on-premise
We want full branding and a custom-built platform	Option C + cloud
Data is extremely sensitive and cannot leave the building	Option A or B + on-premise server
Agents work remotely or from multiple locations	Any option + cloud + VPN
We want the lowest possible ongoing cost long term	On-premise — high upfront, very low monthly. Pays off in 12–18 months
We use Microsoft 365 and need enterprise compliance	Azure cloud — best Microsoft integration and compliance
We want to test before committing to a larger investment	Start with RunPod (~$1.64/hr), validate, then decide

6. Technologies Used

Every component is open-source and runs locally on your server. No subscriptions, no per-message fees, and no vendor lock-in.

Tool	Role	Plain-language explanation
Ollama	LLM runtime	Manages the AI model locally — the engine room that keeps LLaMA running
LLaMA 3.3 70B	The AI model	The brain. 70 billion parameters, created by Meta, open-source. Generates all answers, runs 100% on your hardware
Qdrant	Vector database	Stores your documents as meaning fingerprints. Finds the most relevant content in milliseconds, even with millions of entries
LlamaIndex	RAG coordinator	Connects everything: reads documents, creates meaning fingerprints, talks to Qdrant, assembles the final prompt
FastAPI	Backend server	Lightweight web server that receives questions and returns answers — the bridge between interface and AI model
Open WebUI	Chat interface	The webpage agents visit to chat. Looks like ChatGPT but runs entirely on your server
sentence-transformers	Embedding model	Converts text into numerical meaning fingerprints for semantic search, optimized for English retrieval
PyMuPDF	Document reader	Extracts text from your PDF files when new documents are added to the knowledge base
Docker	Container runtime	Runs components in isolated containers, making deployment and updates significantly simpler

No per-message fees. Once set up, there are no ongoing API costs — only server hardware or rental. At call center query volumes, this is significantly more economical than paying per message to external AI providers.

6.1 Hardware requirements

Component	Minimum	Recommended
GPU VRAM	2× A100 40GB (see note)	2× A100 80GB (see note)
System RAM	64 GB	128 GB
Storage	500 GB SSD	1 TB NVMe SSD
OS	Ubuntu 22.04	Ubuntu 22.04 LTS
Network	Internal only	Internal only — isolated

> **GPU note.** LLaMA 3.3 70B at full precision needs ~140 GB of GPU memory. With Q4 quantization — a compression technique preserving 95%+ of quality — it runs on just 40 GB. A single NVIDIA A100 80GB card is sufficient.

7. Updating the Knowledge Base

Updating the knowledge base is completely independent from the AI model. You never need to retrain or reinstall anything — new content can be added and made available in minutes.

7.1 Adding new documentation pages

— Place the new PDF files in the designated documentation folder on the server
— Ask the technical team to run the indexing process, or add them via remote access

— Done — new content is available immediately, no restart needed

7.2 Adding new solved cases

— Place the new case PDF files in the designated cases folder on the server
— Ask the technical team to run the indexing process, or add them via remote access
— Done — new cases are instantly searchable

> **Automatic updates.** The technical team can set up a process that detects when new files are added and indexes them immediately — keeping the knowledge base always current without any manual steps.

8. Frequently Asked Questions

Does the chatbot send our data to the internet?

No. The AI model, the databases, and the chat interface all operate locally on your server. No agent questions, no case data, and no documentation content is ever transmitted to any external company or AI provider.

What if an agent asks about something not in the documentation?

The system will say clearly that it does not have information on that topic. The AI is instructed to answer only from the provided context — it will not guess or fabricate an answer.

Can multiple agents use it at the same time?

Yes. The system handles concurrent requests simultaneously. Response time may increase slightly under very heavy load, but the system is designed for full call center usage.

How long does the initial setup take?

The documentation manual indexes in a few minutes. The full resolved case library will take several hours during initial setup — a one-time process. After that, adding new content is fast and done without downtime.

What if the answer is wrong?

Occasional misinterpretation happens. The most common cause is unclear or outdated source documentation. Answer quality is directly tied to the quality of the source material. The AI's instructions can also be refined over time to improve accuracy.

Can we add document formats other than PDF?

Yes. The system supports Word documents, plain text files, HTML pages, and several other formats. The document processing component can be extended for additional source types.

Can it integrate with our existing ticketing or CRM system?

Yes, with Option B or C from Section 4. When embedded in your existing application, the AI can connect to your ticketing system or CRM and automatically have context about the current client or case — without the agent typing any of that information manually.